

BME170

Lab-B: PTC Testing Lab

**Single Nucleotide Polymorphism analysis of the PTC
gene**

Assigned Day: Week 9 and 10

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Lab Date: Thursdays, 05/29/25 & 6/05/25

I affirm, under penalty of law, that I completed the entire lab report on my own by following the UCI Academic Integrity Policy at <https://aisc.uci.edu/students/academic-integrity/index.php>.

Hypothesis

The purpose of this experiment is to determine whether I carry one or both alleles of the TAS2R38 gene, which encodes a receptor involved in detecting the bitter compound phenylthiocarbamide (PTC). Based on a qualitative taste test conducted in the lab, I identified myself as a weak taster. Therefore, I hypothesize that I am heterozygous for the TAS2R38 gene, carrying one taster allele and one non-taster allele. To test this hypothesis, I will perform a Polymerase Chain Reaction (PCR) and restriction digest, followed by agarose gel electrophoresis on DNA extracted from my cheek cells.

Methods

Cheek cell samples were collected by rinsing the mouth with a saline solution for 60 seconds. The sample was transferred to a microcentrifuge tube and centrifuged to pellet the cells. After discarding the supernatant, the pellet was resuspended in lysis buffer and incubated at 55°C, followed by 99°C to lyse the cells and release DNA. The lysate was centrifuged again, and the supernatant, containing the extracted DNA, was collected for PCR.

To amplify the region of the *TAS2R38* gene containing the SNP of interest, the DNA sample was combined with a PTC primer mix and a PCR EdvoBead™ containing Taq polymerase and necessary reagents. PCR amplification was performed using a thermal cycler under the following conditions:

- Initial denaturation at 94°C for 4 minutes
- 35 cycles of: 94°C for 30 seconds, 64°C for 30 seconds, 72°C for 30 seconds
- Final extension at 72°C for 5 minutes

After amplification, the PCR product was split into two aliquots. One aliquot was treated with the restriction enzyme HaeIII, which selectively cuts the taster allele at a specific recognition site (GGCC), and incubated at 37°C for 30 minutes. The other aliquot served as an undigested control. Both samples were then loaded into a 2.0% agarose gel alongside a DNA ladder and subjected to electrophoresis. The resulting band patterns were visualized using SYBR® Safe stain and a transilluminator to determine genotype.

Results

Figure 1 presents the PCR amplification curves for Brian's and Landon's DNA samples alongside a positive control. All three samples demonstrated consistent amplification patterns, with Brian's and Landon's Cq values (24.15 and 27.66, respectively) falling within close range of the positive control sample (22.22). This suggests that the DNA concentration and

amplification efficiency were comparable across samples.

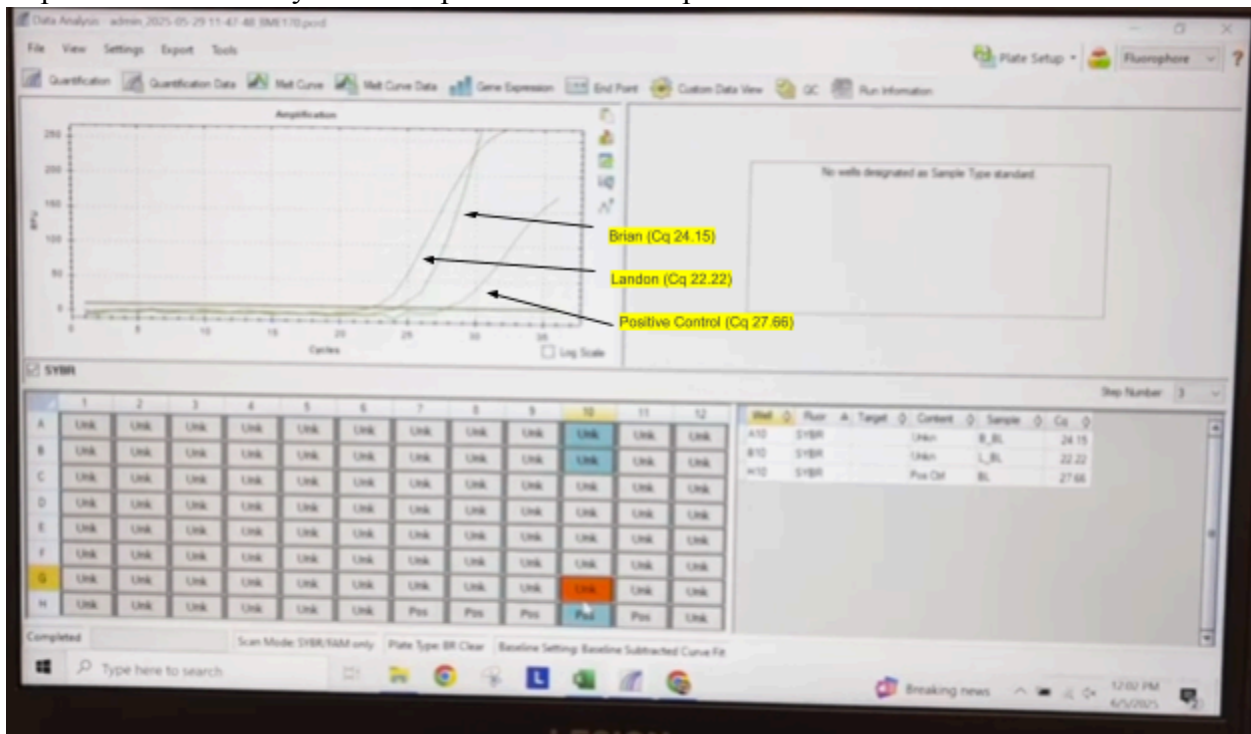


Figure 1: Amplification and Cq Data

Figure 2 illustrates the melting temperature profiles for the same DNA samples. Both Brian’s and Landon’s curves matched the positive control curve’s trajectory, exhibiting similarly defined melt peaks. The consistency in peak shape and temperature indicates that a specific PCR product was generated in each case, supporting the reliability of the amplification results.

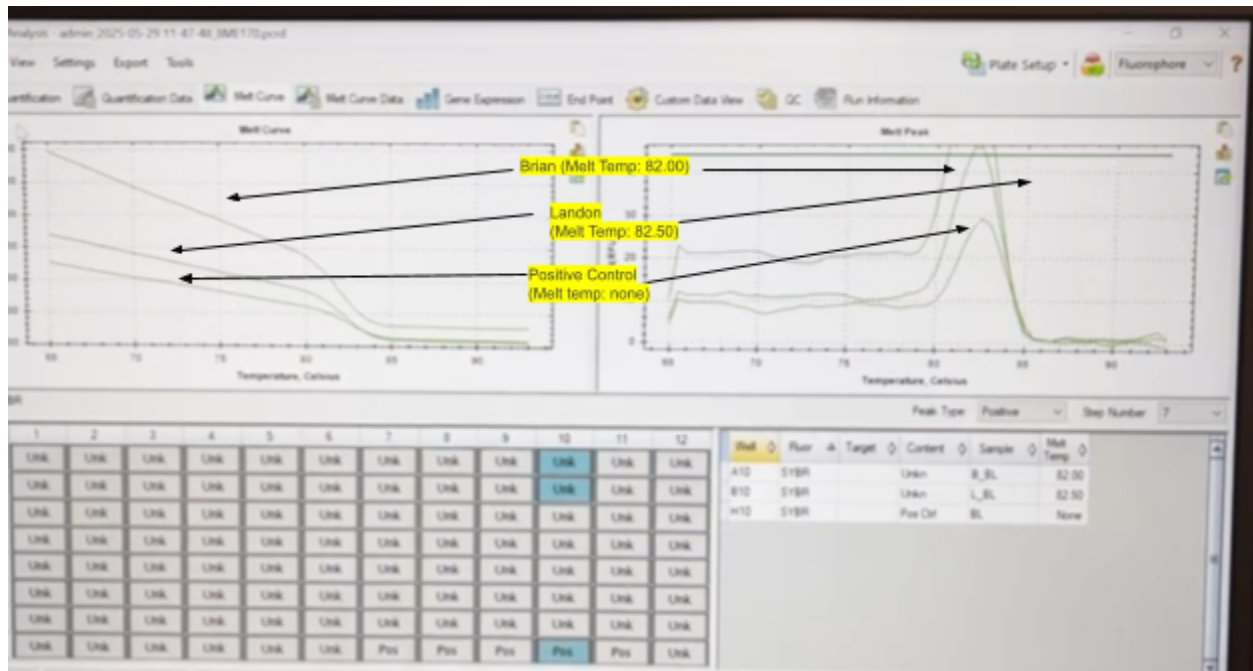


Figure 2: Sample Melt Curve, Melt Peak, and Melt Temp

The final stage of the experiment involved analyzing the PCR products via agarose gel electrophoresis. As shown in Figure 3, DNA samples from Brian and Landon, both in diluted and undiluted forms, were compared alongside a positive control and a DNA ladder. However, due to sample limitations, the volume of DNA loaded was insufficient (less than the ideal 25 μ L), which reduced band intensity and made interpretation more challenging.

Despite this, both Brian's and Landon's data displayed a single, distinct DNA band in their respective wells, consistent with homozygous genotypes. Specifically, each sample showed a single uncut band corresponding to the original 221 bp PCR product, with no evidence of digestion into smaller fragments (e.g., 177 bp and 44 bp). This suggests that both individuals are homozygous for the non-taster allele (tt), which lacks the HaeIII restriction site and remains undigested.

To support this conclusion, we also refer to Figure 4, which shows successful gel electrophoresis results from another lab group. This image illustrates the expected outcomes for digested versus undigested samples. In the unknown samples shown, each digested and undigested pair exhibited the same banding pattern, a single strong band, indicating that the restriction enzyme had no effect. This confirms that all three unknowns are also homozygous recessive (tt).

The DNA ladder provides molecular weight references, while the positive control appears slightly heavier than the unknown samples, consistent with previous melt curve and mass data. Overall, the presence of a single band in all unknown and control samples suggests that none of the tested individuals carried the dominant taster allele.

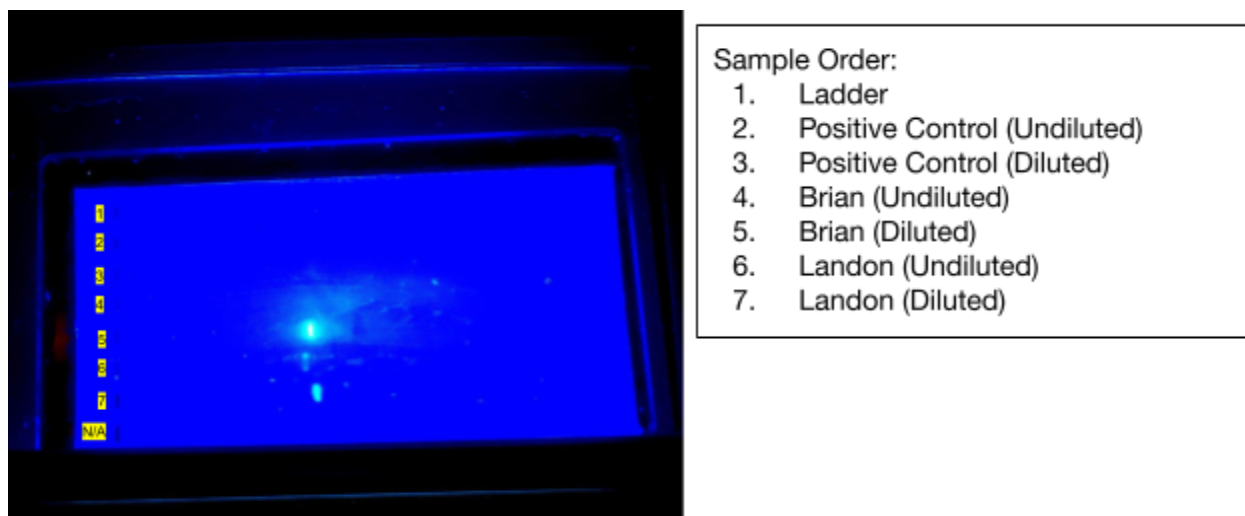


Figure 3: Results of our Agarose Gel Electrophoresis



Figure 4: Successful Results of another Agarose Gel Electrophoresis

Conclusion and Discussion

This experiment aimed to explore the genetic basis of bitter taste perception by analyzing the presence of a single-nucleotide polymorphism (SNP) in the TAS2R38 gene using PCR, restriction digest with HaeIII, and agarose gel electrophoresis. The hypothesis was that I would be heterozygous (Tt), corresponding to a weak taster phenotype. However, the results of gel electrophoresis revealed a single undigested band at 221 bp for both my sample (Brian) and Landon's, indicating that both individuals are homozygous for the non-taster allele (tt).

The absence of restriction enzyme activity in our samples suggests that the recognition site for HaeIII (GGCC) was not present, consistent with the non-taster allele. While a weak taster phenotype might suggest heterozygosity (Tt), the genetic data showed no digestion, implying that either phenotypic interpretation was subjective or environmental/epigenetic factors may have influenced perceived bitterness. This highlights a limitation in relying solely on taste tests to infer genotype, as taste sensitivity can vary due to non-genetic factors such as age, oral health,

or perception thresholds.

Furthermore, despite some faint bands and limited sample volumes in our gel (Figure 3), the clarity and consistency of the undigested pattern across all lanes, including comparison to the clearer results in Figure 4, support the conclusion that the individuals tested were homozygous non-tasters. The use of a DNA ladder and control lanes helped validate band size and confirmed that amplification and loading were successful.

This experiment demonstrates the utility of PCR and restriction fragment length polymorphism (RFLP) analysis in detecting SNPs that correlate with phenotypic traits. While taste tests provide a quick, qualitative assessment, molecular techniques offer a more accurate and objective method for determining genotype. Our findings confirmed that both participants lacked the dominant taster allele and were genetically non-tasters, emphasizing the importance of integrating both phenotypic and genotypic data for comprehensive analysis.